

ICT — Digital Society 2: Key Notes

資訊科技「Digital Society 2」重點筆記 · Worksheet due 14 Oct · 6B ICT (Mr. Man)

Built from the actual "Digital Society 2" worksheet in Google Classroom

1. What Is an Algorithm? 什麼是演算法？

Algorithm = a set of **step-by-step instructions** to complete a task.

演算法 = 一步一步的指令，用來完成一項任務。

- Example (Task 1): "Get ready for school" — 1. Turn off alarm → 2. Brush teeth → 3. Get dressed → ...
- Good algorithms are **clear, ordered, and complete** — anyone following them gets the same result.

2. Algorithms in Digitalisation 數碼化中的演算法

Algorithm 演算法	Example 例子	What it does 功能
Pathfinding 路徑搜尋	Maps suggests the fastest route	Finds the best/shortest path between two places
Recommendation 推薦	Netflix recommends shows you might like	Suggests things based on what you watched/liked before
Ranking 排序	Google shows the most relevant search results first	Orders results by how relevant or important they are
Classification 分類	Email filters spam messages	Sorts things into groups (spam / not spam)

★ **Match them like this:** Maps → Pathfinding · Netflix → Recommendation · Google search → Ranking · Spam filter → Classification. For "advantages and disadvantages", think: **faster & personal BUT can be wrong, biased, or leak privacy.**

3. When Digitalisation Goes Wrong 數碼化出錯時

Digital tools help us, but they can also cause **human errors** — mistakes people make because they trust the technology too much.

Digitalisation 數碼化	Possible human error 可能的人為錯誤
GPS navigation reduces travel stress	Following GPS blindly into a wrong or unsafe road
Automated reminders for homework	Forgetting to check — "the app didn't remind me"

Cloud storage keeps files safe	Losing everything if the account is hacked or the internet fails
Online banking is convenient	Falling for phishing scams / typing the wrong account number
E-commerce simplifies shopping	Impulse buying; buying fakes by mistake
Digital attendance systems	Marked absent because the card/scan failed
Faster communication (emails, IM)	Sending to the wrong person; misunderstandings without face-to-face tone

4. Digitalisation at School — Before & After 學校數碼化：前後對比

Activity	Before Digitalisation 以前	After Digitalisation 現在
Homework	Paper worksheets, hand in by hand	Google Classroom, submit online
Permission slips	Paper reply slips, parents sign	E-notices, e-signatures
Meetings	Everyone in one room	Video calls (Zoom/Meet)
Attendance	Teacher calls names	Scan cards / digital roll
Government	Queue at offices with paper forms	Online services and e-forms
Finance	Cash and bank books	E-payment and online banking
Healthcare	Paper records, long waits	E-health records, online booking

5. Recap: Digitalisation 1 重溫 Digitalisation 1

- **Digitalised** = the **process** is done digitally (e.g. submitting homework on Classroom).
- **Digitised** = the **item** is turned into digital form (e.g. a paper worksheet scanned to PDF).
- Reasons for school digitalisation: faster, saves paper, easier to share and store.
- Positive impacts: convenient, quick, eco-friendly. Negative impacts: screen time, tech failures, less face-to-face contact.

★ **Worksheet tips (due 14 Oct):** ① Task 1 — write at least 8 ordered steps for "get ready for school". ② Task 4 — invent your own daily-life algorithm (e.g. "make breakfast", "pack school bag") with numbered steps. ③ Task 5 — one clear before/after contrast per row. ④ Remember to bring your Digitalisation 1 homework too (Mr. Man's note).